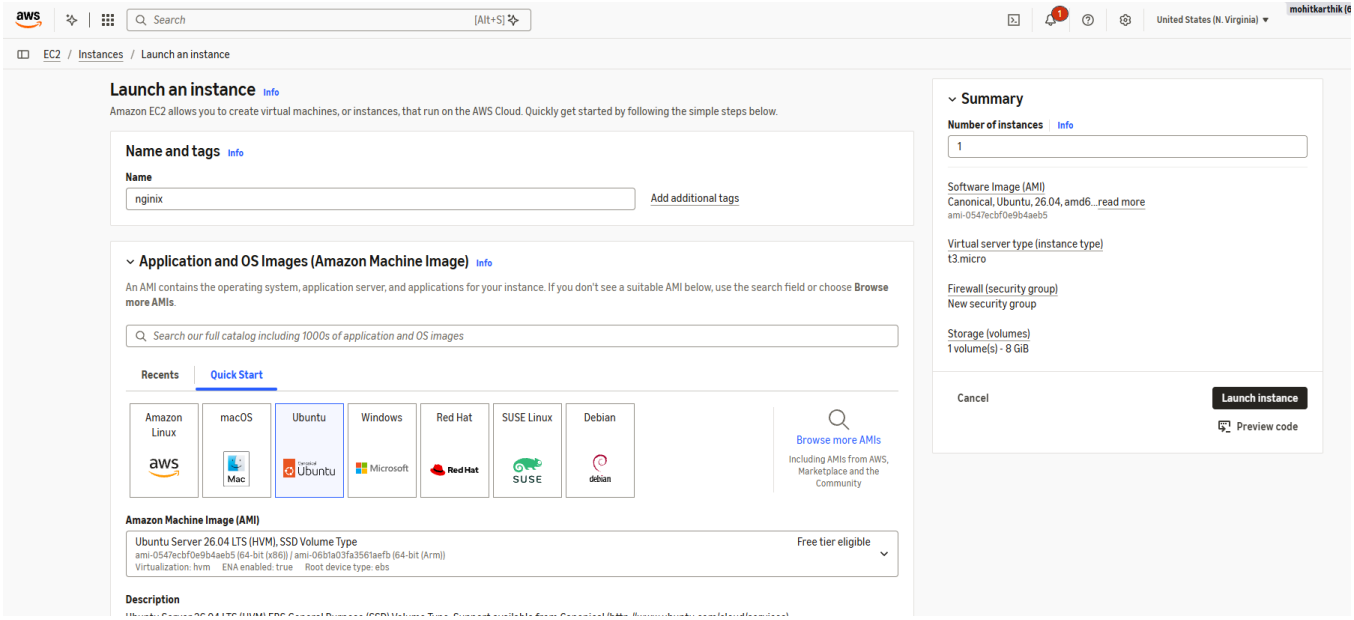
SKILL 3

Launch an Ubuntu EC2 Instance and Host a Website Using Nginx

Aim: To launch an Ubuntu virtual machine on Amazon EC2, install the Nginx web server using user data, and verify the website through a browser.

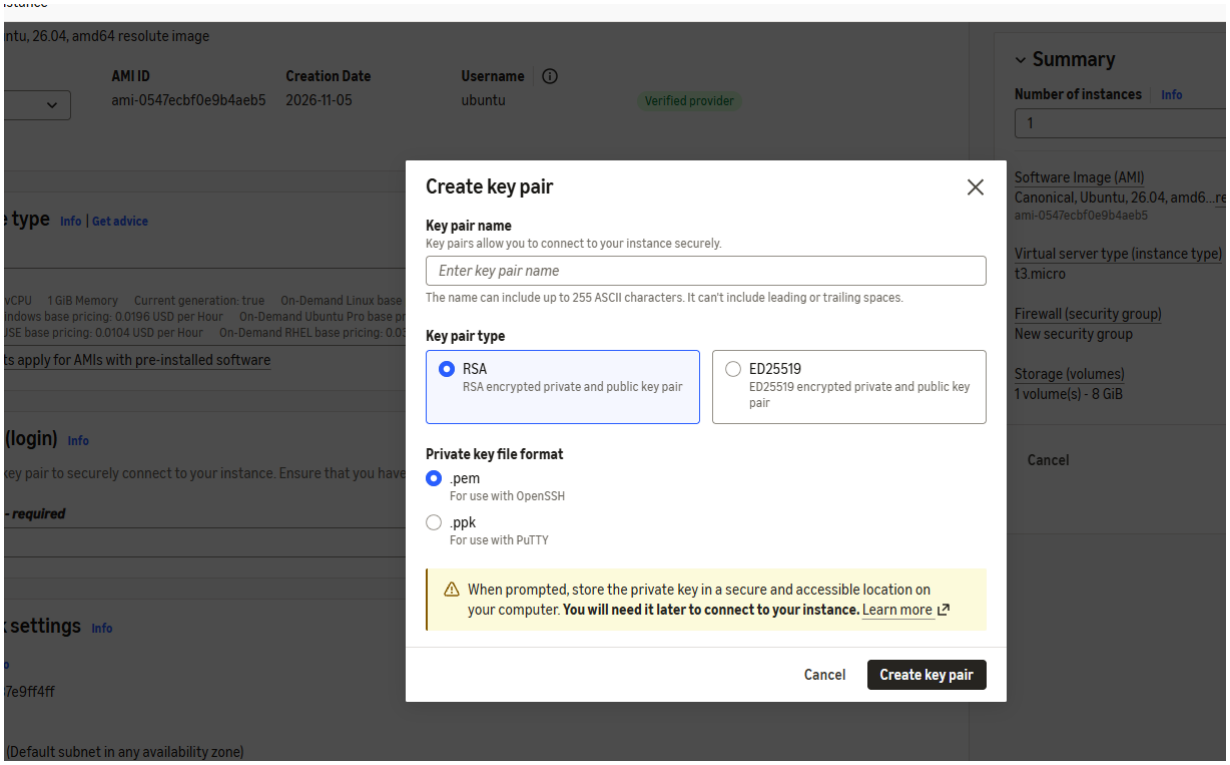
Step 1: Configure the EC2 instance

Open Amazon EC2 and choose Launch instance. Enter the instance name as **nginx** and select Ubuntu Server as the operating system image (AMI).



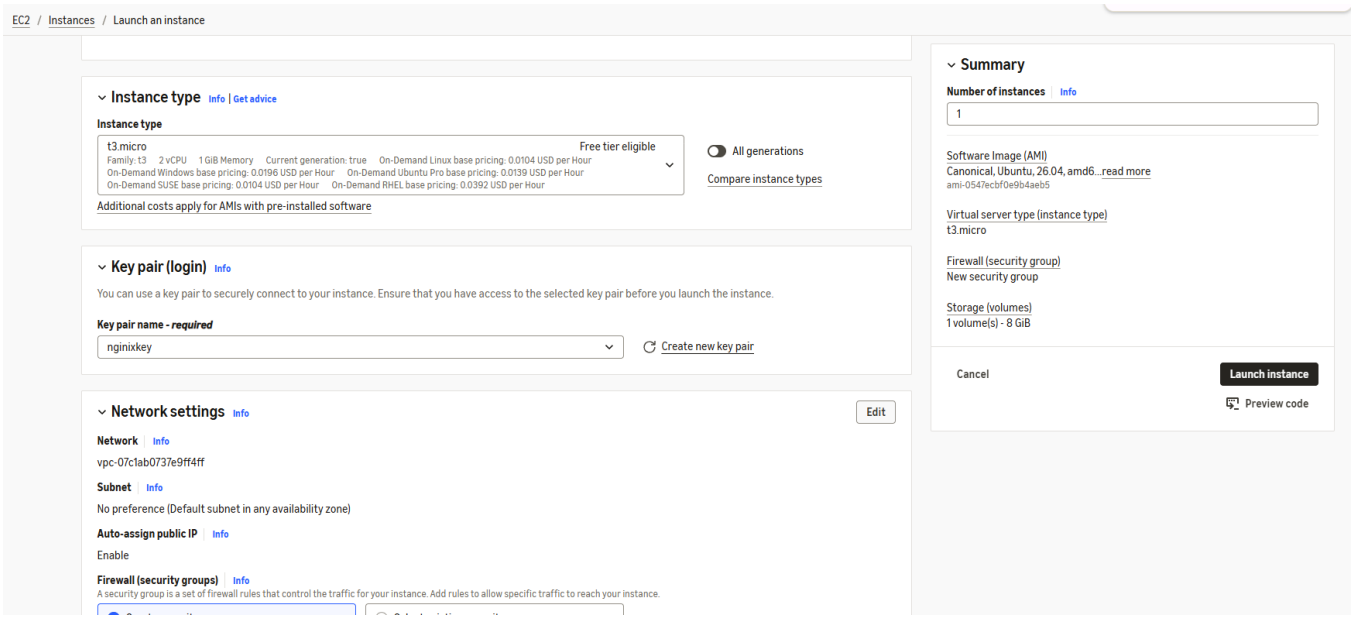
Step 2: Create a key pair

Create a key pair named **nginxkey**. The key pair is used for secure access to the virtual machine. Save the downloaded private key file in a safe place.



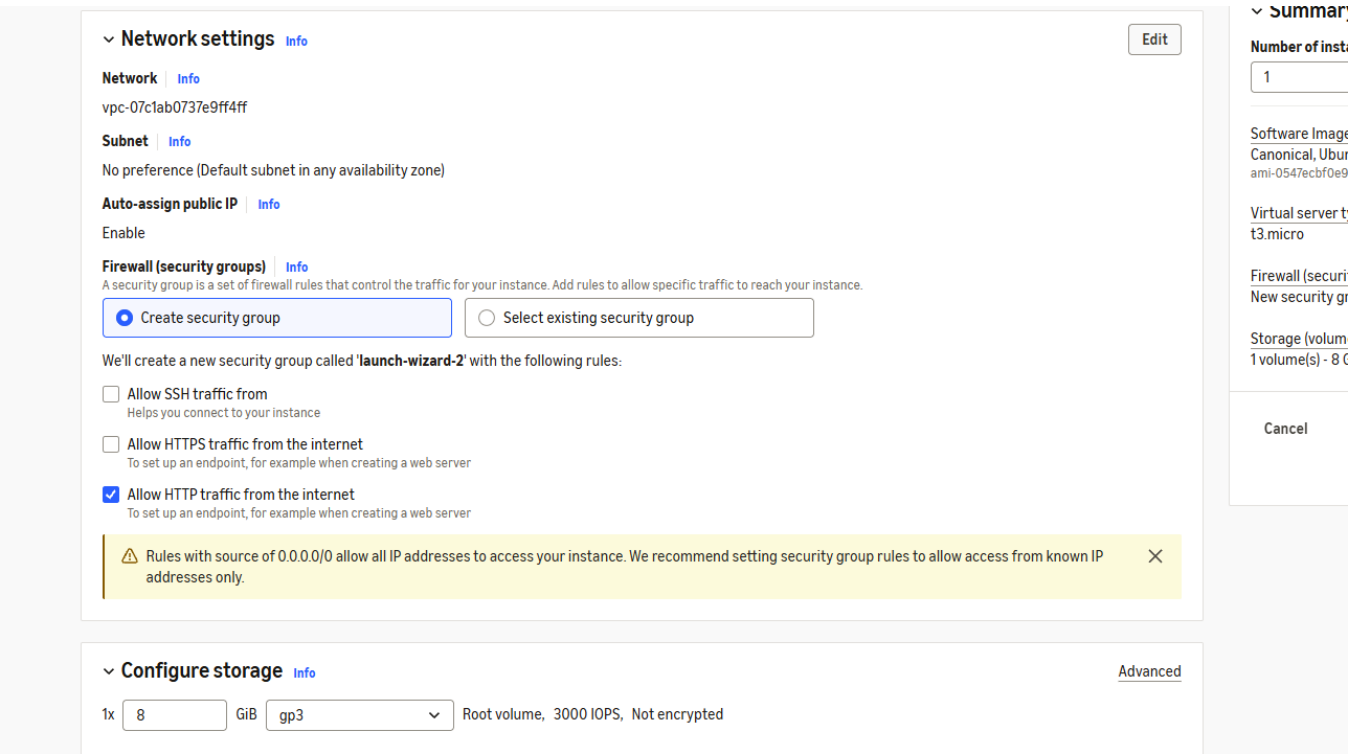
Step 3: Select instance type and key pair

Choose the **t3.micro** instance type and select the newly created **nginxkey** key pair. This configuration is shown in the launch settings.



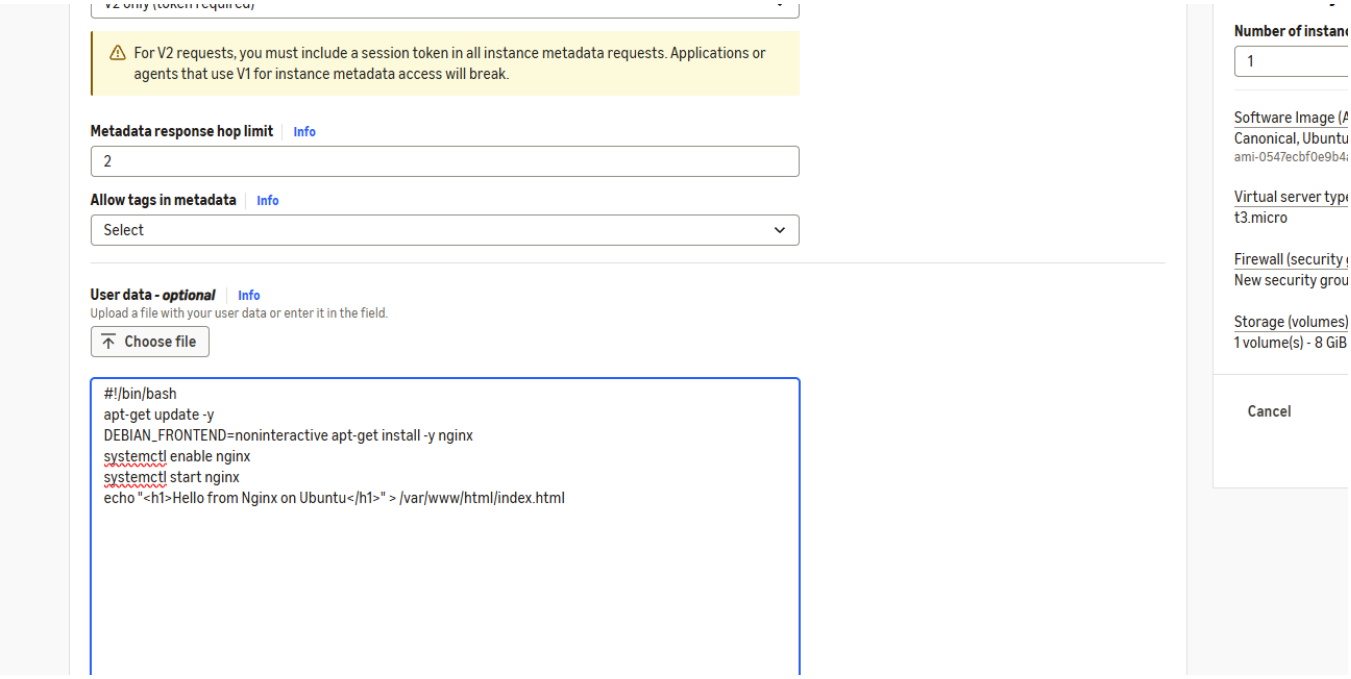
Step 4: Configure network and storage

Keep the selected VPC and default subnet. Enable auto-assign public IP and allow HTTP traffic from the internet so that the web page can be opened in a browser. The screenshot shows an 8 GiB gp3 volume.



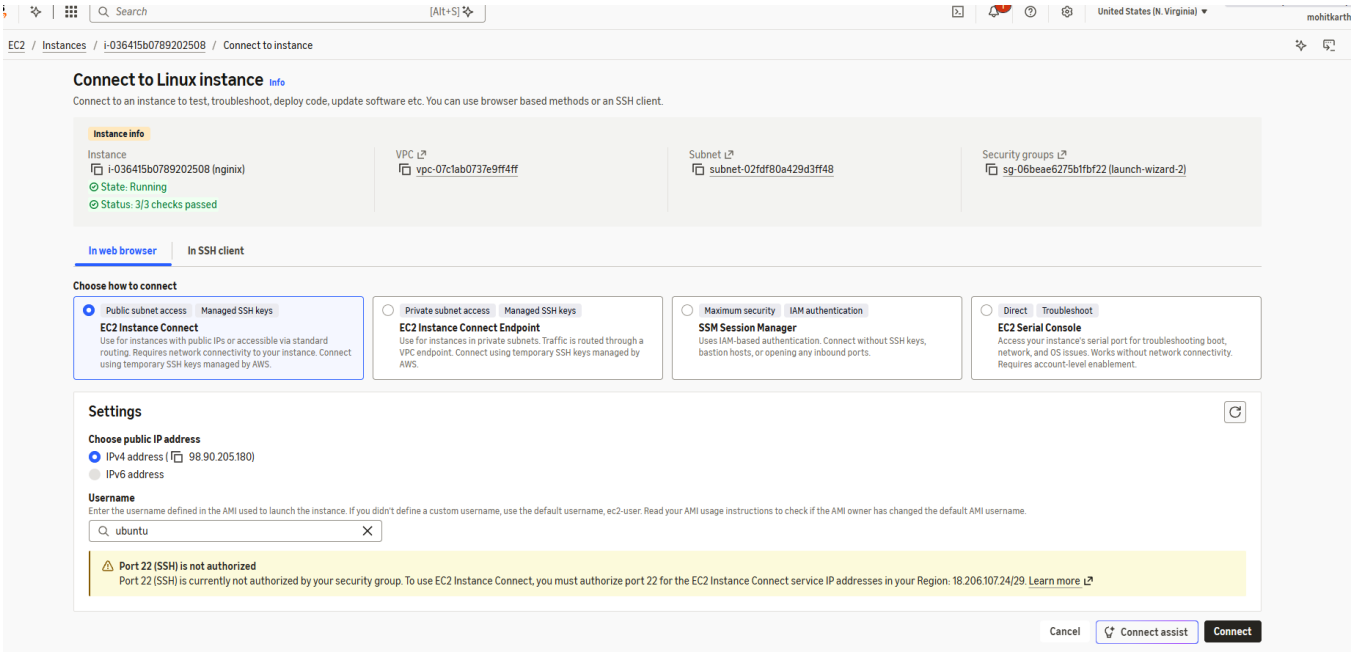
Step 5: Add the installation commands

In the User data field, enter a startup script that updates Ubuntu, installs Nginx, enables and starts the Nginx service, and writes a simple welcome message to the web page.



Step 6: Connect to the running instance

After launching, open the instance connection page. Confirm that the instance is running and its status checks have passed. The screenshot notes that SSH port 22 is not authorized for the selected connection method; browser access to the HTTP page is demonstrated next.



Step 7: Verify the Nginx web page

Open the instance's public IPv4 address in a web browser. The message **Hello from Nginx on Ubuntu** confirms that Nginx is installed and serving the page.



Hello from Nginx on Ubuntu

Result: An Ubuntu EC2 instance was launched and configured with Nginx. The browser displayed the custom message “Hello from Nginx on Ubuntu,” confirming that the web server was working.